

Chaitanya Bojji

Frisco, TX

224-704-8400 | cbojji2003@gmail.com

EDUCATION

The University of Texas at Dallas

Computer Engineering Major

Richardson, TX

Aug. 2021 – May 2025

EXPERIENCE

Ansys

Jan. 2023 – April 2023, Aug. 2023 - Dec. 2023

Geometry Integration Design Intern

Remote

- Developed Python wrappers for the geometry model API using Boost.Python, CMake, and Linux, enabling cross-platform debugging application development.
- Integrated 200+ unit tests using C# and .NET framework within CAD software codebase, improving model transfer reliability and preventing post-release defects.
- Applied C++ and object oriented programming to enhance backend geometry model API, adding complex model structure support, 10+ new features for existing models, and expanded model compatibility.
- Implemented GRPC protocols between internal services, streamlining system modularity and data exchanges

Motio Inc.

Sep. 2022 - Dec. 2022

Quality Assurance Intern

Remote

- Contributed to the investigation of 50+ software bugs by reproducing issues, analyzing logs, and controlling test environments to support developer debugging and resolution.
- Performed feature and regression testing in agile sprints, validating edge cases and contributing to sprint planning for full QA coverage.
- Updated QA documentation by adding 10+ new test cases for assigned features, increasing scenario coverage and improving clarity for future regression testing.

Spin Servers

Oct. 2021 – March 2022

Research Intern

Remote

- Explored architecture designs for a 100-node Linux-based cloud cluster with Redis, providing insights on scalability, redundancy, and fault tolerance.
- Documented the setup and testing of a Redis-based cluster, configuring nodes and verifying fail-over behavior, while using a GUI tool for real-time monitoring of test clusters.
- Executed stress tests on 20-node clusters by simulating heavy data transfers, measuring upload/download performance under peak load, and identifying areas for scalability improvements.

PROJECTS

AI Facial Recognition Locking System | *Raspberry Pi, OpenCV, Flask, Python, AWS*

- Built an IOT embedded system with a solenoid lock, 12V relay module, door reed sensor, and Raspberry PI using GPIO for automated access control.
- Programmed Python scripts with the OpenCV framework to use a machine learning model that authenticates authorized users and triggers door lock/unlock mechanisms.
- Utilized AWS DynamoDB to store sensor data and generate visualizations of authorized vs. unauthorized access attempts identified by the ML model's training algorithm.

Covid ML Classification Model | *Python, Pandas, Numpy, Scikit-Learn, Matplotlib, XGBoost*

- Preprocessed tabular classification data using Pandas, NumPy, and Scikit-Learn for data cleaning, label encoding, feature selection, and train-test split.
- Tuned hyperparameters on a XGBoost classifier using RandomizedSearchCV and early stopping, optimizing custom decision thresholds to improve F1-score balance across classes in an imbalanced classification task.
- Visualized classification performance using Matplotlib and Seaborn, including confusion matrices, precision-recall curves, and feature importance plots to interpret model behavior.

TECHNICAL SKILLS

Languages and Frameworks: Java, Python, C++, C#, GRPC, Boost-Python, SQL, OpenCV, Scikit

Technologies: Git, Microsoft Office Suite, Visual Studio, Tableau, Cad, CMake, Cadence Design Suite, AWS

Relevant Coursework: Machine Learning, Artificial Intelligence, Database Systems, Data Structures & Algorithms